

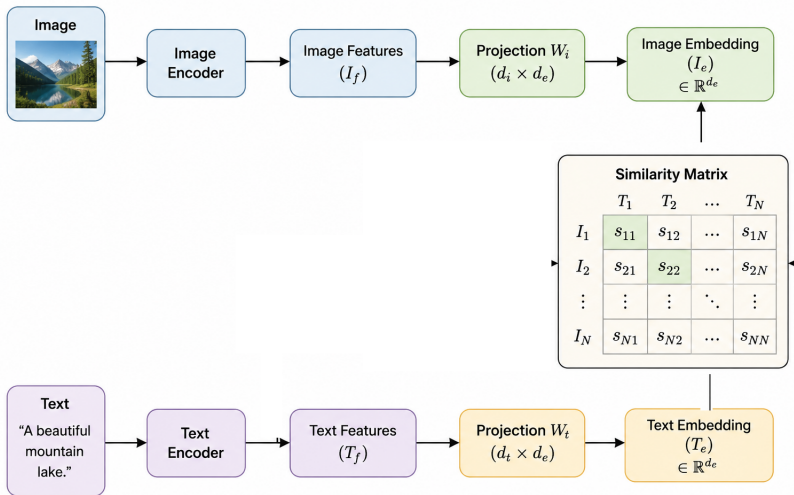
CLIP Explained with a Toy Example

Mathematical Formulation and Intuition

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CLIP



What is CLIP?

CLIP (Contrastive Language-Image Pretraining) learns a shared embedding space between images and text.

Key Idea

Matching image-text pairs should have high similarity, while mismatched pairs should have low similarity.

Examples:

- Image of a dog $\leftrightarrow$ “dog”
- Image of a cat $\leftrightarrow$ “cat”
- Image of a car $\leftrightarrow$ “car”

Enables:

- Zero-shot classification
- Image retrieval
- Vision-language understanding

Toy Example

Suppose we have three image-caption pairs.

Image	Text
Dog image	“dog”
Cat image	“cat”
Car image	“car”

Batch size:

$$N = 3$$

Positive pairs correspond to diagonal entries.

Step 1: Image Encoder

Assume the image encoder produces

$$I = \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 2 \end{bmatrix}$$

where

$$I_i = f(I)$$

Image	Feature Vector
Dog	[2,1]
Cat	[1,3]
Car	[4,2]

Step 2: Text Encoder

Suppose text features are

$$T = \begin{bmatrix} 1.8 & 1.1 \\ 1.2 & 2.8 \\ 3.7 & 1.9 \end{bmatrix}$$

where

$$T_i = g(T)$$

Text	Feature Vector
Dog	[1.8,1.1]
Cat	[1.2,2.8]
Car	[3.7,1.9]

Step 3: Feature Normalization

CLIP performs L2 normalization:

$$z_I = \frac{I}{\|I\|}$$

For the dog image

$$I = [2, 1]$$

Norm:

$$\|I\| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

Normalized vector:

$$z_{dog}^{img} = [0.894, 0.447]$$

Similarly,

$$z_{dog}^{txt} = [0.853, 0.521]$$

Step 4: Cosine Similarity Matrix

Cosine similarity:

$$s_{ij} = z_i^T z_j$$

Similarity matrix:

$$S = \begin{bmatrix} 0.996 & 0.746 & 0.998 \\ 0.654 & 0.998 & 0.712 \\ 0.997 & 0.738 & 0.999 \end{bmatrix}$$

Diagonal entries correspond to positive pairs.

Step 5: Temperature Scaling

CLIP scales similarities using temperature τ :

$$L = \frac{S}{\tau}$$

Typically

$$\tau = 0.07$$

Example:

$$\frac{0.996}{0.07} = 14.23$$

Temperature sharpens the probability distribution.

Step 6: Softmax Probability

For the dog image, the scores are

[14.23, 10.66, 14.26]

Probability of the correct caption:

$$P(\text{dog}|\text{image}) = \frac{\exp(14.23)}{\exp(14.23) + \exp(10.66) + \exp(14.26)}$$

Image-to-Text Contrastive Loss

$$L_{i2t} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ij}/\tau)} \right)$$

Objective:

- Increase similarity of positive pairs.
- Decrease similarity of negative pairs.

Text-to-Image Contrastive Loss

$$L_{t2i} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{\exp(s_{ii}/\tau)}{\sum_j \exp(s_{ji}/\tau)} \right)$$

Both modalities are optimized simultaneously.

$$L = \frac{L_{i2t} + L_{t2i}}{2}$$

Symmetric Training

Image $\rightarrow$ Text

Text $\rightarrow$ Image

This produces a common embedding space.

Bidirectional Learning

Image-to-Text:

Dog Image $\rightarrow$ {dog, cat, car}

Text-to-Image:

“dog” $\rightarrow$ {Dog Image, Cat Image, Car Image}

CLIP learns both directions simultaneously.

Zero-Shot Classification

Given an unseen image

Image $\rightarrow ?$

Candidate labels:

- “dog”
- “cat”
- “horse”
- “car”

Prediction:

$$\hat{y} = \arg \max_j s_{ij}$$

No task-specific retraining is required.

Key Takeaways

- 1 CLIP learns a shared image-text embedding space.
- 2 Uses cosine similarity and contrastive learning.
- 3 Trains image-to-text and text-to-image jointly.
- 4 Enables zero-shot recognition.
- 5 Forms the foundation of modern vision-language models.

Thank You